

APPLICATION OF THE KRUSKAL-WALLIS TEST TO COVID-19 VACCINATION DATA - A CASE STUDY AT UMaT

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ABSTRACT

This project examined COVID-19 vaccination among students of the University of Mines and Technology (UMaT). I used survey data to look at vaccination patterns, attitudes and perceptions among the students. The Kruskal-Wallis test was used to examine whether vaccination rates differed across academic-year groups.

1. INTRODUCTION

The COVID-19 pandemic led to widespread vaccination programmes and also raised questions about vaccine uptake and people's attitudes towards vaccination. I chose UMaT students as the focus of this study and used statistical analysis to examine the vaccination information collected from them. The Kruskal-Wallis test was selected because the study compared vaccination rates across more than two academic-year groups.

2. OBJECTIVES

The study was carried out with the following objectives:

- To apply the Kruskal-Wallis test to COVID-19 vaccination data across academic-year groups.
- To examine vaccination patterns among UMaT students.
- To provide information about students' vaccination behaviour, attitudes and perceptions.

3. METHODOLOGY

3.1 Data Acquisition and Collection

I collected the data using a structured questionnaire. The project included procedures for distributing the questionnaire, selecting participants through random sampling, obtaining informed consent and considering the sample size. The questionnaire covered demographic information and vaccination-related questions, including vaccination status, vaccine type, and number of doses, attitudes and perceptions.

3.2 Data Preparation and Analysis

After collecting the responses, I prepared and cleaned the data before carrying out the analysis. The analysis involved ranking the observations, calculating sums of ranks and the grand total, and producing descriptive summaries. I then applied the Kruskal-Wallis test and considered its assumptions, hypothesis, significance level, test statistic and interpretation. The project also discussed the use of post-hoc tests.

4. DATA ANALYSIS AND RESULTS

The study examined vaccination responses in relation to academic year and gender. It also looked at vaccine type, vaccine hesitancy, knowledge about the vaccine, booster uptake and the number of doses received.

MEASURE	RESULT
Kruskal -Wallis H	3.00
Degree Of Freedom	3
P- Value	0.392
Significance level	0.05

Measure Kruskal-Wallis H Degrees of freedom p-value Result 3.00 3 0.392 Significance level 0.05 The Kruskal-Wallis test gave an H value of 3.00 with 3 degrees of freedom and a p-value of 0.392. Since the p-value was greater than the 0.05 significance level, the study reported no statistically significant difference in vaccination rates among the academic-year groups.

5. CONCLUSION AND RECOMMENDATIONS

The study did not find a statistically significant difference in vaccination rates among the academic-year groups. The results also showed the importance of looking beyond vaccination status alone and considering students' knowledge, attitudes, hesitancy and other vaccination-related experiences. The project recommended improving awareness about COVID-19 vaccination, addressing concerns that may contribute to hesitancy, building trust and providing useful vaccination information to students.

6. RESEARCH PROCESS AND SKILLS DEVELOPED

Working on this project gave me experience with the main stages of a quantitative research study. I collected questionnaire data, prepared and cleaned the responses, organized the observations, carried out statistical calculations and interpreted the results. I also gained experience with sampling, informed consent, descriptive statistics, non-parametric testing, hypothesis testing and interpreting statistical results. Most importantly, the project showed me how the mathematics and statistics I studied could be used to investigate a real problem.

7. RELEVANCE TO FURTHER STUDY

This project was one of the experiences that increased my interest in statistics and data analysis. Working with real survey data and applying the Kruskal-Wallis test made me interested in developing stronger skills in statistical computing, programming and data-driven problem solving. These interests have influenced my decision to pursue further study in areas involving statistics, data science and computational research.

SOURCE Original source: *Application of Kruskal-Wallis Test to COVID-19 Vaccination Data - A Case Study at UMaT*, BSc Mathematics project report, University of Mines and Technology, Tarkwa, Ghana, September 2023